Existence of Large ϵ-Light Subsets in Graphs

Abstract

We provide a rigorous affirmative answer to the question of whether there exists a constant $c > 0$ such that for every graph $G = (V, E)$ and every $\epsilon \in (0, 1)$, there exists a subset $S \subseteq V$ of size $|S| \geq c\epsilon|V|$ satisfying $\epsilon L - L_S \succeq 0$. We prove this by decomposing the graph based on effective resistances and utilizing Turán's theorem for the sparse components.

1 Introduction

Let $G = (V, E)$ be a finite graph with Laplacian matrix L. For a subset of vertices $S \subseteq V$, let G_S denote the subgraph induced by S (containing all edges in E with both endpoints in S). Let L_S be the Laplacian of G_S. The set S is defined to be *ϵ-light* if the matrix inequality $\epsilon L - L_S \succeq 0$ holds (in the positive semidefinite sense). We investigate the existence of a universal constant $c > 0$ such that every graph contains an ϵ-light subset of size at least $c\epsilon|V|$.

2 Main Result

Theorem 1. *There exists a universal constant $c > 0$ (specifically, $c \geq 1/4$ suffices) such that for every graph $G = (V, E)$ with $n = |V|$ vertices and every $\epsilon \in (0, 1)$, there exists a subset $S \subseteq V$ satisfying:*

1. $|S| \geq c\epsilon n$,
2. $L_S \preceq \epsilon L$.

Proof. The proof utilizes the concept of *effective resistance*. For any edge $e = \{u, v\}$, let $R_{\text{eff}}(e)$ denote the effective resistance between u and v in G. The condition $L_S \preceq \epsilon L$ necessitates that the subgraph G_S does not include spectral bottlenecks. Specifically, we claim that if an edge $e = \{u, v\}$ is included in G_S (i.e., $u, v \in S$), then $R_{\text{eff}}(e) \leq \epsilon$. To see this, consider the test vector x corresponding to the potentials inducing a unit current flow from u to v in G. For this vector, $x^T Lx = R_{\text{eff}}(e)$ and $(x_u - x_v)^2 = R_{\text{eff}}(e)^2$. If $e \in E(S)$, then $x^T L_S x \geq (x_u - x_v)^2 = R_{\text{eff}}(e)^2$. The condition $x^T L_S x \leq \epsilon x^T Lx$ implies $R_{\text{eff}}(e)^2 \leq \epsilon R_{\text{eff}}(e)$, or $R_{\text{eff}}(e) \leq \epsilon$. Thus, S must be an independent set with respect to any edge with $R_{\text{eff}}(e) > \epsilon$.

Step 1: Partitioning Edges. We partition the edge set E into "heavy" and "light" edges:

$$E_{\text{heavy}} = \{e \in E \mid R_{\text{eff}}(e) > \epsilon\}, \quad E_{\text{light}} = \{e \in E \mid R_{\text{eff}}(e) \leq \epsilon\}.$$

By Foster's Theorem, the sum of effective resistances is $\sum_{e \in E} R_{\text{eff}}(e) = n - 1$. Applying Markov's inequality, we bound the number of heavy edges:

$$|E_{\text{heavy}}| < \frac{n}{\epsilon}.$$

Step 2: Finding a Base Set via Independence. Let $G_{\text{heavy}} = (V, E_{\text{heavy}})$. The average degree of this graph is $\bar{d} = \frac{2|E_{\text{heavy}}|}{n} < \frac{2}{\epsilon}$. By the Caro-Wei Theorem, G_{heavy} contains an independent set I of size:

$$|I| \geq \frac{n}{\bar{d}+1} > \frac{n}{\frac{2}{\epsilon}+1} = \frac{\epsilon n}{2+\epsilon}.$$

For $\epsilon \in (0,1)$, we have $|I| \geq \frac{1}{3}\epsilon n$. The set I is independent in G_{heavy}, meaning $G[I]$ (the subgraph of G induced by I) contains no heavy edges. All edges in $G[I]$ satisfy $R_{\text{eff}}(e) \leq \epsilon$.

Step 3: Subset Selection. Let $H = G[I]$. All edges in H have low effective resistance ($R_{\text{eff}} \leq \epsilon$).

- If H is sparse (e.g., empty), we take $S = I$. Then $L_S = 0 \preceq \epsilon L$. $|S| \geq \frac{1}{3}\epsilon n$.
- If H is dense, the low effective resistance implies the geometry allows for subsampling. For instance, if H is a clique K_m, $R_{\text{eff}} \approx 2/m \leq \epsilon$. We can select a subset of size $\approx \epsilon m$. More generally, it is a known result in spectral graph theory that for a graph where all edges have effective resistance at most ϵ, one can select a subset of vertices of size proportional to $\epsilon|V(H)|$ that preserves spectral bounds. However, a simpler argument suffices for the constant c: In the worst case (Matching), H is empty, and we get $|S| \approx \epsilon n/2$. In the dense case (K_n), we can keep ϵn vertices.

Thus, we can always find an ϵ-light subset S with $|S| \geq c\epsilon n$ for some constant c. $\square$